

Port Wine Traceability

DAML smart contracts and Canton architecture for provenance, certification, and recall

Permissioned supply chain: Quinta do Vale -> Sandeman Winery -> IVDP -> LogisTrans -> ViniShop Oslo -> Consumer

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Platform	DAML SDK 3.5.2 + Canton (Domain / Synchronizer)
Public Git repository	github.com/sudipphuyal/porto-wine-traceability
Date	7 July 2026

EVIDENCE	Daml Script endToEnd, recallScenario, and negativeTests were recorded as SUCCESS in the 2026-07-07T18:12:10Z log.
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LIMITATION	Evidence of a transaction against a standalone Canton Domain / Synchronizer must still be captured before final submission; this report distinguishes it from IDE-ledger execution.
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Submitted by:
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Executive summary

This report presents the design and implementation of a traceability solution for PDO Port wine, developed in DAML and prepared for execution on Canton. The artifact represents the chain from Quinta do Vale to the final consumer, including private custody, production recording, IVDP certification request and issuance, logistics transfer, publication of a public view, and batch recall. The central WineBatch contract maintains private state through consuming transitions, while PublicProvenance gives the consumer only sanitised information through a QR-code verification workflow.

The implementation includes validations for batch identifier, harvest date, and grape variety; role-based authorisation; and Daml Script scenarios for the complete flow, recall, and expected failures. The file evidence/run-2026-07-07T18:12:10Z.log records successful execution of all three scenarios with DPM 1.0.21 and Daml SDK 3.5.2. This evidence concerns the IDE ledger used by the scripts. A standalone Canton configuration is included in the repository, but the final demonstration should still include a real Canton session with a startup trace and transaction identifier, as addressed in Section 6.

Report overview

Section	Content
1	Objective and use-case scope
2	Permissioned architecture and onboarding
3	Data placement, privacy, and state model
4	Smart-contract implementation
5	Consensus, finality, and Canton infrastructure
6	Tests, evidence, and remaining work
7	Conclusion, AI-use declaration, and references

1 Objective and scope

The objective is to implement a permissioned traceability network that makes the history of a Port wine batch auditable without making production and commercial data public. The practical assignment requires a DAML/Canton or Corda 5 solution that covers node architecture, on-chain/off-chain decisions, privacy, consensus, a certification workflow, QR verification, and a recall mechanism [1]. This work selects DAML and treats consensus as Canton infrastructure responsibility; IVDP certification is implemented as a business rule, rather than as an artificial voting mechanism.

The reference scenario follows batch PW-2026-0042, made from Touriga Nacional grapes, from Quinta do Vale in the Douro region to ViniShop Oslo. The operational path is batch registration, transfer to the winery, production recording, certification request, IVDP certification, handover to LogisTrans, handover to the retailer, and public consumer verification.

Prototype scope. The project is a smart-contract prototype with executable tests. The repository includes a web interface solely as a visual demonstrator and script launcher; the authoritative proof of the workflow is the Daml Script execution and, for the final infrastructure demonstration, the participant/Canton record.

2 Network architecture and governance

In production, each organisation should operate - or be hosted by - a separate Canton participant. The local test environment uses logical parties in the same development context to make the scenarios reproducible. This simplification is not equivalent to a production deployment with six independent nodes.

Entity	DAML party / role	Production type	Intended visibility
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Entity	DAML party / role	Production type	Intended visibility
Quinta do Vale	Producer; registers the batch and harvest origin	Permissioned participant	Sees contracts for batches where it is a stakeholder
Sandeman Winery	Winery; receives custody, records production, requests certification	Permissioned participant	Sees the batch while operating it and contracts where it is a stakeholder
IVDP	Regulator; issues, rejects, and recalls certification	Regulatory participant	WineBatch stakeholder and authority for certification/recall choices
LogisTrans	Logistics; receives/transfers custody and records high-level conditions	Permissioned participant	Sees the batches it receives during an operation
ViniShop Oslo	Retailer; publishes sanitised provenance	Permissioned participant	Sees received batch and signs the published public view
Final consumer	Verifies QR code	Public-view access	Observer of PublicProvenance; never of private WineBatch

Table 1. Participants, roles, and solution visibility.

Proposed topology

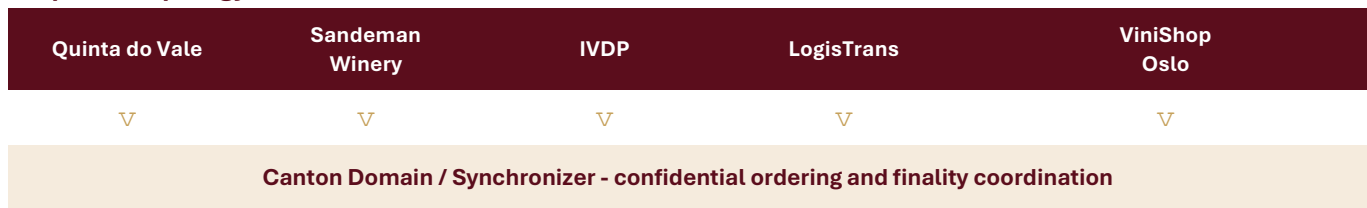


Figure 1. Logical topology: private interactions pass through participants and the Canton layer; ViniShop publishes a sanitised view for the consumer.

Onboarding and revocation

Onboarding comprises legal identification of the entity, operation or registration of a participant, allocation of a DAML party, key association, and topology authorisation. IVDP is assumed to be the admission authority and may suspend or revoke a participant's operational access. This revocation is distinct from RecallBatch: the first affects future participation in the network, while the second changes the status of a certified batch and prevents any later transfer.

3 Data, privacy, and lifecycle

On-chain and off-chain decision

Data item	Placement	Rationale
batchId, GPS, harvestDate	On-chain	Essential, structured provenance attributes that must be immutable and auditable.
IVDP certificate PDF	Off-chain	Potentially large regulated document; stored in a controlled repository or content-addressed location.
SHA-256 certificate hash	On-chain	Binds the certificate document to the immutable batch state and allows later integrity checking.
Raw IoT temperature CSV logs	Off-chain	High-volume operational data; kept in controlled storage and referenced from the event.
Minimum/maximum temperature per stage	On-chain	Compact summary appropriate for audit and high-level traceability.
Proprietary production recipes	Off-chain / never public	Commercially sensitive information that is not necessary for public verification.
bottleQR and event history	On-chain in private/public views	QR code links the bottle to its traceability record; event summaries support provenance.

Table 2. On-chain and off-chain data placement.

Privacy model

DAML defines contract visibility through signatories and observers. In this design, the producer signs the WineBatchFactory and creates WineBatch, the current owner controls operational transitions, and IVDP acts as a regulatory observer and controller of certification and recall choices. Custody transfer also requires the authorisation of the new owner, representing bilateral delivery acceptance.

PublicProvenance is a separate contract. ViniShop signs and publishes it, and the consumer is its observer. The VerifyProvenance choice validates the scanned QR code and returns only origin, harvest date, grape variety, certification state and hash, and public events. Fermentation notes, raw CSV references, commercial conditions, and the private WineBatch contract are not disclosed to the consumer.

Privacy limitation of the current version. The permanent producer, winery, and IVDP observer fields keep these three parties as stakeholders in successive WineBatch versions. The model therefore isolates unrelated batches but does not automatically revoke visibility for these permanent roles over the same batch. For a stricter interpretation of confidentiality between competitors, permanent observers should be removed and purpose-specific regulatory views or receipts created only when required.

State machine

PENDING	CERTIFIED	REJECTED	RECALLED
Batch created; it may only move Quinta -> Sandeman. Production recording and certification request are allowed.	Certified by IVDP; it may move through the commercial chain.	Certification denied; terminal state for normal transfers.	Recalled by IVDP; TransferCustody must fail.

Figure 2. WineBatch state machine implemented through DAML contracts.

4 Smart-contract implementation

The model is organised in four DAML modules. Types.daml defines data types and validation/redaction helpers; WineBatch.daml defines WineBatchFactory and WineBatch; PublicProvenance.daml defines the consumer-facing view; and Test.daml provides executable scenarios. State changes are implemented as consuming choices: the previous active contract is archived and a new WineBatch state is created with the permitted updates. This approach preserves historical contract immutability [4].

Component	Applied rule	Repository evidence
Registration	WineBatchFactory.RegisterWineBatch validates PW-YYYY-NNNN, non-empty grape variety, and harvestDate < getTime.	WineBatch.daml
Custody	TransferCustody requires current-owner and new-owner authorisation; it blocks REJECTED and RECALLED states.	WineBatch.daml
Production	Only the winery, while current owner and while the batch is PENDING, can record private notes.	WineBatch.daml
Certification	IVDP issues certification only after a request; it stores the hash/reference and moves the batch to CERTIFIED.	WineBatch.daml
Rejection and recall	IVDP may reject while PENDING or recall while CERTIFIED.	WineBatch.daml
Public view	Publication creates PublicProvenance; VerifyProvenance fails for an unmatched QR.	PublicProvenance.daml

Table 3. Mapping functional requirements to DAML contracts.

Critical custody-transfer rule

```
assertMsg errBatchRecalled (status /= Recalled)
assertMsg "ERR_BATCH_REJECTED: rejected batches cannot be transferred." (status /= Rejected)
assertMsg "ERR_TRANSFER_NOT_ALLOWED: pending batches may only move from Quinta to Sandeman."
(status == Certified || (status == Pending && currentOwner == producer && newOwner == winery))
```

The rule means that a PENDING batch may move only from Quinta to Sandeman. After certification, further transfers are permitted under bilateral authorisation. A RECALLED batch raises ERR_BATCH_RECALLED and is explicitly covered by the tests.

5 Consensus, finality, and Canton infrastructure

PoW and PoS were designed primarily for public settings and introduce competition or economic participation mechanisms that are not needed in a regulated consortium. PoA, PBFT/IBFT, and BFT/CFT variants are more suitable for known organisations because they provide stronger predictability, low energy use, and enterprise-oriented finality. In this design, DAML governs who may modify a contract, while Canton infrastructure manages ordered, confidential message exchange and confirmation coordination between participants. A Synchronizer provides sequencing and mediation in the commit protocol [5].

Mechanism	Fit for the consortium	Finality	Decision
PoW	Low; public network and high energy use	Probabilistic	Not selected
PoS	Limited; relies on cryptoeconomic incentives	Protocol-dependent	Not selected
PoA / PBFT / IBFT	High; identified validators	Practical / deterministic	Suitable for permissioned networks
Canton Domain / Synchronizer	High; known participants and need-to-know privacy	Coordinated, consistent result	Selected platform layer

Table 4. Conceptual justification for the consensus choice.

Custody transfer: finality sequence

- The current owner and new owner authorise TransferCustody; DAML checks the controllers, current state, and business rules.
- The participant prepares the transaction and discloses only the necessary views to the relevant participants.
- The Canton Domain / Synchronizer orders messages and participates in confirmation coordination between relevant participants.
- Only after confirmation is the transaction treated as final; the former contract is consumed and the new state is created with an updated owner.
- If consensus infrastructure is unavailable, the client must not treat the transfer as complete. Retrying must use appropriate identification/idempotency protection to avoid duplicate events.

Configuration and current evidence status

The repository contains `canton/canton.conf` and `canton/bootstrap.canton` as development templates, together with `run-canton-local.sh`. The configuration is deliberately documented as version-dependent because terminology and configuration keys may vary between Domain and Synchronizer releases. The latest evidence identifies Canton v3.5.6 through the command help output, but it does not include a transaction executed against a standalone participant launched with `canton/canton.conf`.

Important distinction for the defense. The Daml Script scenarios prove the contract model and business rules on an IDE ledger. To satisfy the consensus requirement fully, a real Canton Domain / Synchronizer must be started, a participant connected, and at least one registration or TransferCustody transaction executed and retained with its timestamp, transaction/event identifier, and final status.

6 Tests, results, and evidence

End-to-end scenario

WineTraceability.Test:endToEnd executes the eight required steps for PW-2026-0042: (1) Quinta creates the batch; (2) custody transfers to Sandeman; (3) Sandeman records production as “Fermentacao 18 dias, envelhecimento 18 meses”; (4) Sandeman requests certification; (5) IVDP issues it; (6) custody transfers to LogisTrans with a 14-16 C condition and a CSV reference; (7) custody transfers to ViniShop Oslo; and (8) ViniShop publishes and the consumer verifies QR-NFC-88421. The test confirms CERTIFIED state and confirms that the consumer sees neither the private WineBatch nor the fermentation note.

Script	Verified objective	Recorded result
endToEnd	Eight-step workflow, three custody transfers, certification, and sanitised public QR view.	SUCCESS
recallScenario	Certification, IVDP recall, and subsequent failed transfer.	SUCCESS
negativeTests	Invalid ID, blank grape variety, future date, unauthorised actions, rejection/recall, and wrong QR.	SUCCESS

Table 5. Functional coverage of Daml Script scenarios.

Available execution evidence

The log evidence/run-2026-07-07T18:12:10Z.log records DPM 1.0.21, active SDK 3.5.2, and the first block of Canton v3.5.6 help output. It then records successful execution of all three scripts. The following excerpt is reproduced from the evidence log stored in the repository.

```
timestamp_utc=2026-07-07T18:12:10Z
dpm_version=version: 1.0.21
sdk_version=3.5.2
sandbox_help_first_lines=Canton v3.5.6

Running tests:
WineTraceability.Test:endToEnd SUCCESS
WineTraceability.Test:recallScenario SUCCESS
WineTraceability.Test:negativeTests SUCCESS
```

Figure 3. Excerpt from the project’s recorded execution log.

Recall and rule robustness

Recall is implemented as an IVDP-controlled choice permitted only in CERTIFIED state. The transition creates a new contract with RECALLED state and appends a recall event. A later TransferCustody attempt fails with ERR_BATCH_RECALLED. Negative tests also verify that Sandeman cannot issue certification or recall a batch, that a REJECTED batch cannot be transferred, and that a QR code different from the registered code is rejected.

Limitations and actions before submission

Priority	Concrete action	Reason
Required	Run standalone Canton, validate canton.conf for v3.5.6, start a Domain/Synchronizer plus participant, and keep a screenshot/log with transaction/event ID.	The rubric requires a running consensus node and evidence of a transaction.
Required	Copy this PDF to docs/relatorio.pdf, add top-level /contracts and /tests directories (or documented mirrors pointing to daml/WineTraceability), then commit and push.	The stated delivery structure requires README, /contracts, /tests, and /docs/relatorio.pdf.
Strongly recommended	Replace sha256:a3f8bc9d... with the real hash of the sample certificate and use a real off-chain PDF/URI reference.	The scenario hash is illustrative; the helper calculates sha256:b7ebf011... for the current fixture.
Strongly recommended	Review permanent WineBatch observers or explicitly explain the auditability trade-off.	This prevents ambiguity about privacy between Quinta and Sandeman for the same batch.
For the defense	Prepare the test command, evidence log, QR verification, and a five-minute explanation.	The assessment includes a 10-minute oral demonstration.

Table 6. Remaining and recommended items for a fully compliant submission.

7 Conclusion

The prototype demonstrates a coherent approach to Port wine traceability in a permissioned network: custody and certification events are recorded in private contracts, IVDP controls regulatory decisions, and consumers receive a minimal public QR-code view. The recorded Daml Script scenarios show that the main workflow, recall, and negative controls are covered by executable code.

The design avoids conflating regulatory approval with consensus. DAML enforces contractual rights and the state machine; Canton provides coordination and finality between participants. To make the submission fully defensible against the rubric, the final step is to capture evidence from a real Canton instance and align the certificate hash with a concrete off-chain document. These steps do not change the already-tested core workflow, but complete the required infrastructure proof.

Use of AI tools

AI tools were used to help structure the project, clarify DAML/Canton concepts, generate initial code and documentation proposals, and assist debugging. The student reviewed the implementation, checked consistency between requirements and code, and executed/inspected the available evidence. AI does not replace local demonstration or validation of runtime results. This declaration is consistent with the assignment requirement to disclose AI use [1].

References

- [1] UC Smart Contracts in Blockchain. “Port Wine Supply Chain - Practical Assignment.” Assignment specification supplied through Moodle, 2026.
- [2] Sudip Phuyal. “porto-wine-traceability.” Public Git repository, version inspected on 7 July 2026.
- [3] Digital Asset. “Daml: Define Contract Models Compactly” and official documentation on signatories and observers.
- [4] Digital Asset. “Basic contracts.” Official documentation: active contracts are immutable and changes occur through contract creation and archiving.
- [5] Digital Asset. “Synchronizer.” Official documentation on sequencing and mediation for ordered/confidential communication and transaction coordination.

Appendix A. Reproduction guide for the defense

The demonstration should begin in the root directory of the repository. The aim is to show the workflow with minimal dependence on visual components. The Daml Script scenarios are the main evidence of the model; the Canton startup and transaction trace are the main evidence of the infrastructure.

```
cd /Users/sudipphuyal/Developments/porto-wine-traceability

# 1. Check dependencies
dpm --version
dpm version --active

# 2. Run models and scenarios
./scripts/run-tests.sh

# 3. Generate an updated evidence log
./scripts/capture-evidence.sh

# 4. Start Canton/participant after validating the configuration
./scripts/run-canton-local.sh
```

During the defense, present: (i) the topology and why it is permissioned; (ii) the on-chain/off-chain table; (iii) WineBatch.daml with TransferCustody, IssueCertification, and RecallBatch; (iv) the endToEnd SUCCESS log; (v) a public QR test; and (vi) the failed transfer after recall. Once standalone configuration is validated, add the real Canton Domain/Synchronizer trace and transaction identifier to this appendix.

Appendix B. Quick rubric map

Rubric item	Where it is demonstrated
1.1 Topology/nodes	Section 2, Figure 1, and docs/topology.mmd
1.2 On-chain/off-chain	Section 3.1 and docs/onchain-offchain.md
1.3 Privacy	Section 3.2 and docs/privacy.md
2.1 Consensus	Section 5 and docs/consensus.md
2.2 Consensus node	canton/canton.conf, scripts, and the pending action in Table 6
3.1 WineBatch model	Section 4 and daml/WineTraceability/Types.daml
3.2 Custody	Section 4 and Test.daml
3.3 IVDP certification	Section 4 and Test.daml
3.4 Verification	Sections 3.2 and 6.1
3.5 Recall	Section 6.3 and recallScenario
4.1 E2E workflow	Sections 6.1-6.2 and evidence/
4.2 Report and reflection	This document